

2025_HL60_KO_supercharged_construct_data

June 8, 2025

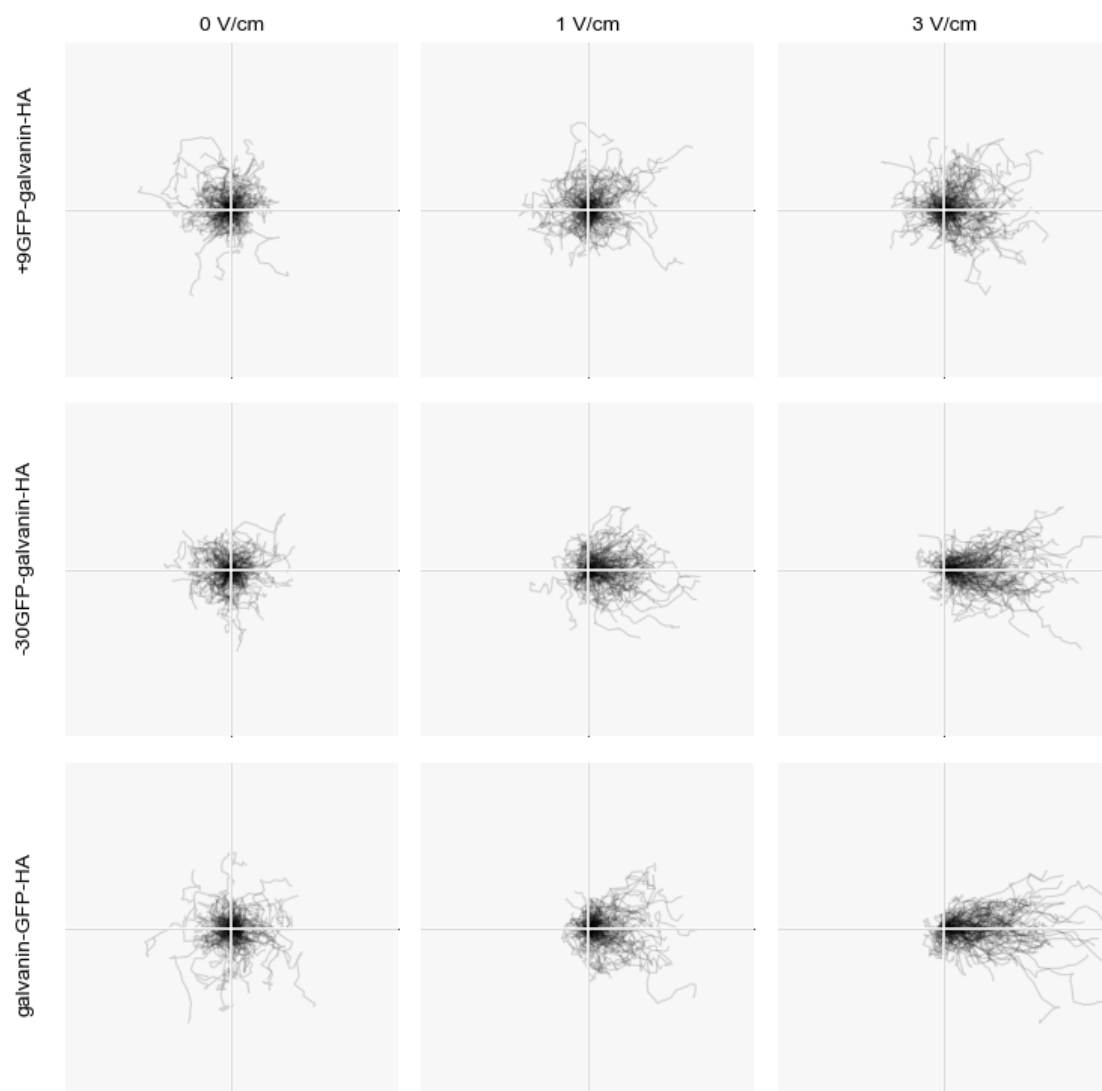
[4]:

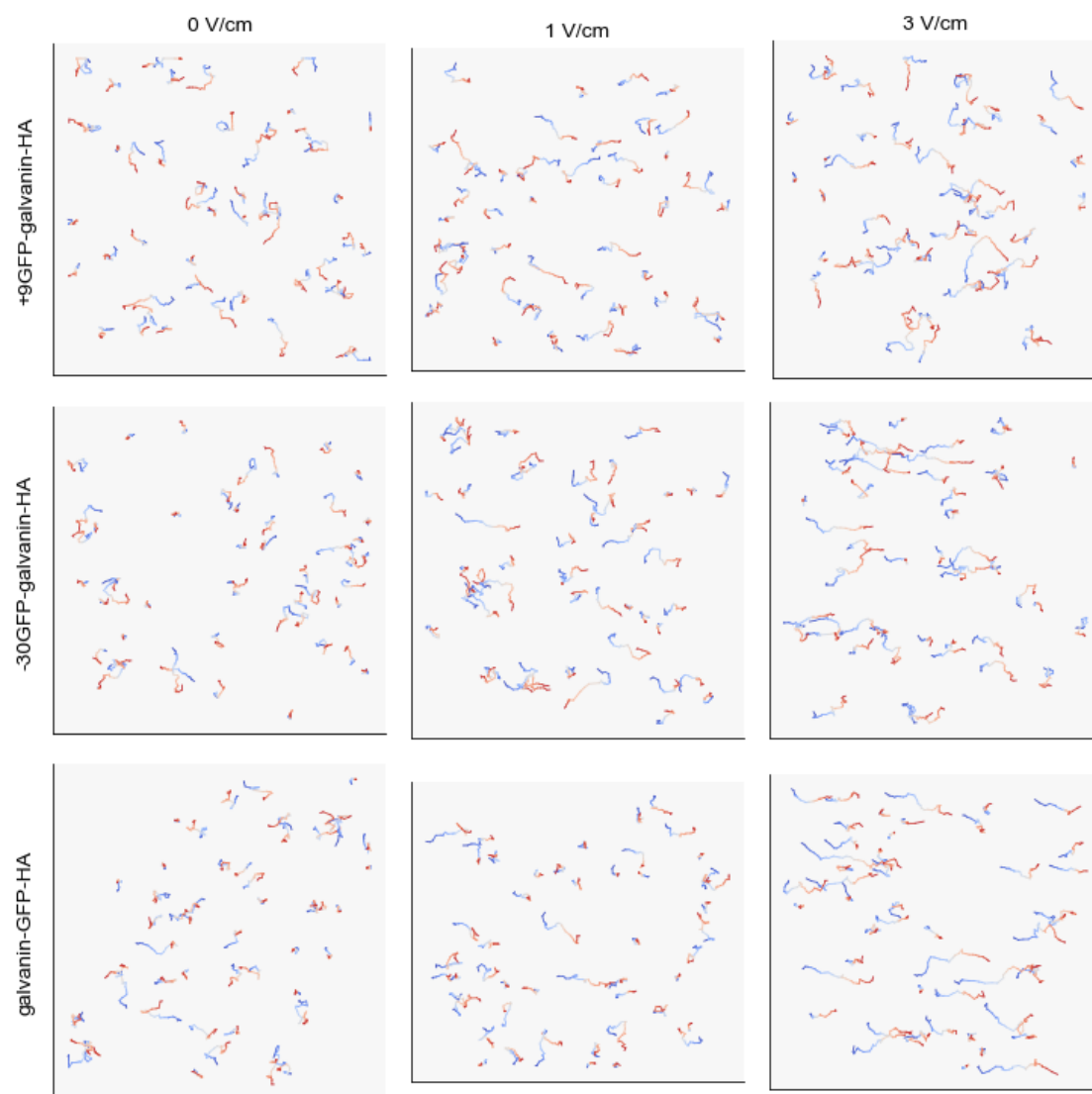
	celltype	E_V_cm	date	trial	user	cell	\
0	dHL60TMEM154KO-clone1+9GFP-galvanin-HA	0	20240515	1	HK	3	
1	dHL60TMEM154KO-clone1+9GFP-galvanin-HA	0	20240515	1	HK	3	
2	dHL60TMEM154KO-clone1+9GFP-galvanin-HA	0	20240515	1	HK	3	
3	dHL60TMEM154KO-clone1+9GFP-galvanin-HA	0	20240515	1	HK	3	
4	dHL60TMEM154KO-clone1+9GFP-galvanin-HA	0	20240515	1	HK	3	

	frame	x	y	z
0	0	252.243925	360.130815	220.702794
1	1	255.418158	363.218407	218.694373
2	2	256.874769	369.624201	216.593420
3	3	257.306670	373.108395	219.479155
4	4	257.538843	373.167138	222.743826

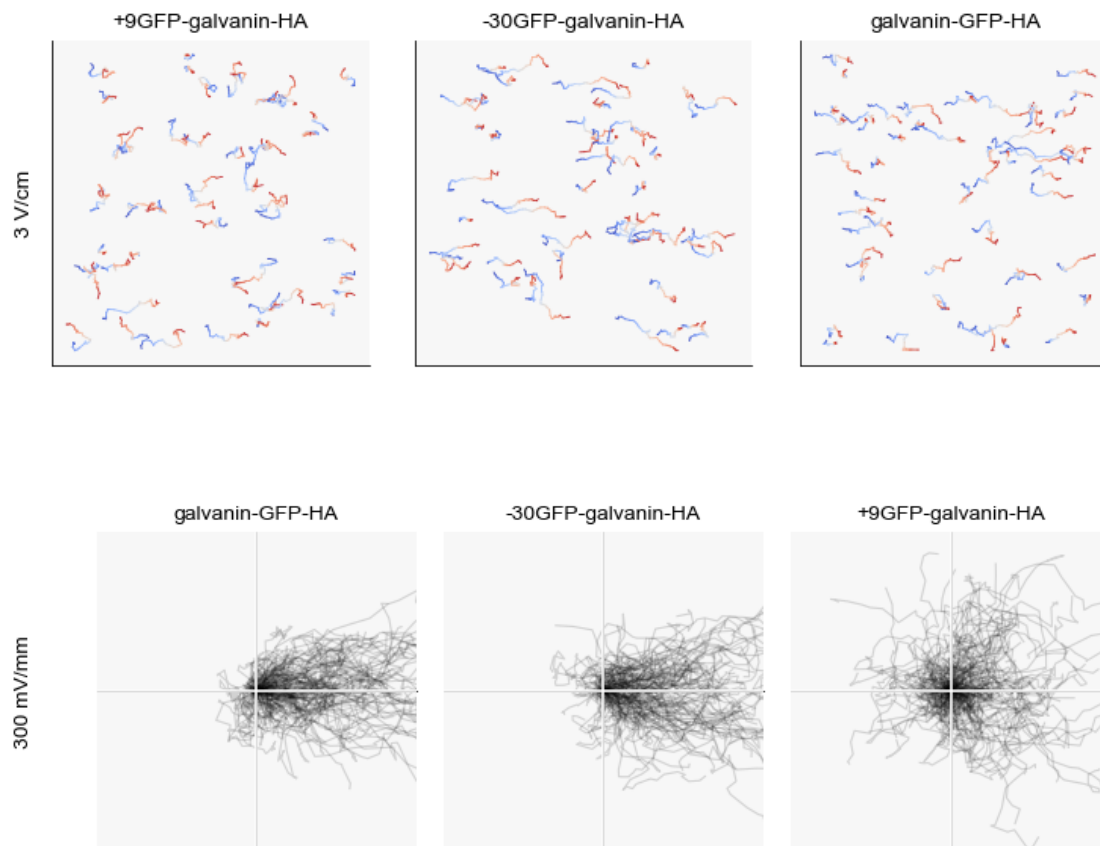
1 Plotting of HL-60 cell lines expressing ‘super-charged’ constructs

1.1 Tracks

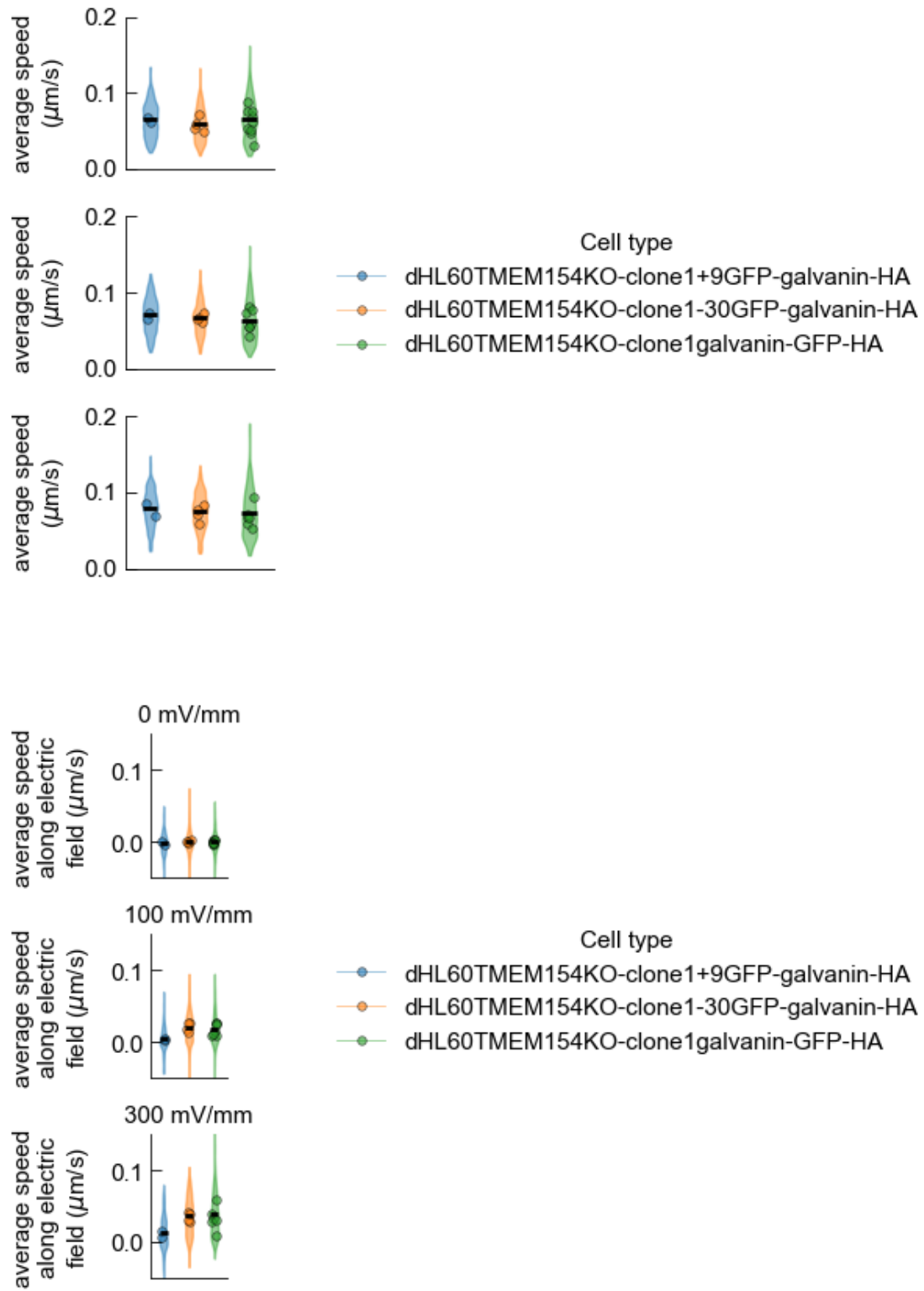




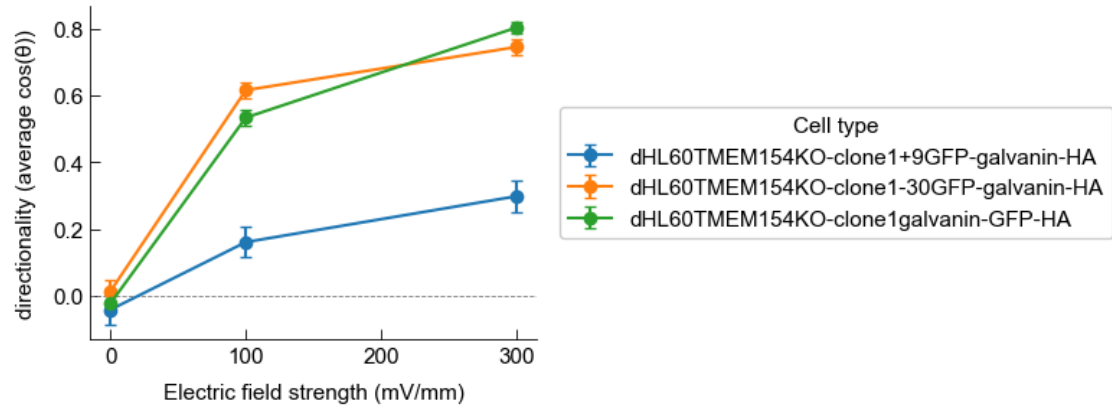
1.2 Here are only plots of 300 mV/mm condition only - probably more useful for main text figures.



2 Migration speed related

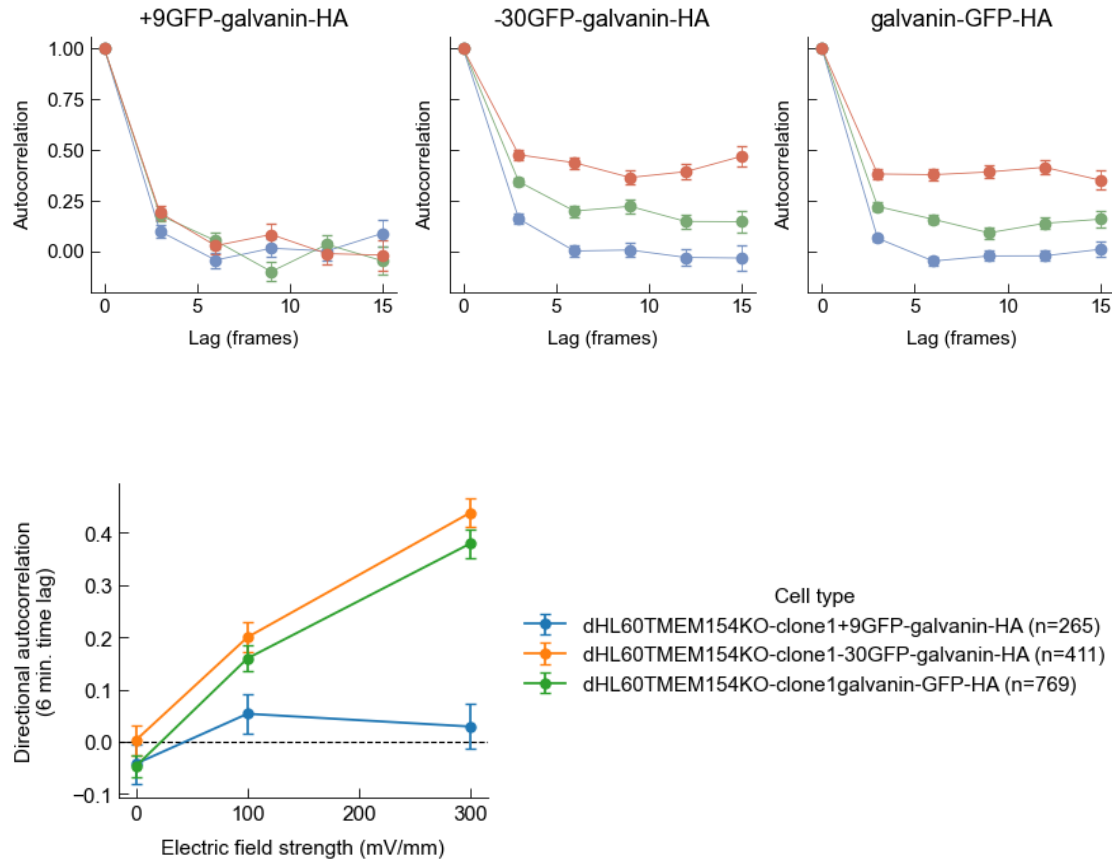


3 Directionality (average cosine theta)



4 Autocorrelation Analysis

This uses cosine theta between trajectory vector and electric field (the regular approach we've been taking).



4.1 Autocorrelation, but relative to 0 mV/mm

Here we plot relative to the 0 mV/mm condition to try to account for residual persistence in the absence of an electric field. (i.e. subtraction of autocorrelation value from the 0 mV/mm condition).

